ENTERPRISE JIRA SPRINT BACKLOG • COMPLETE 124+ COMMIT TRACEABILITY

AutoCare AI: Complete 138-Ticket Jira Agile Backlog

124 Completed Git Commits Mapped (Sprints 1-4) + Future Big Data Transformation Backlog (Sprints 5-6)

Project Key & Repo:	Key: **AUT** | Remote: github.com/marketbul/autocare-ai.git
Completed Work (Done):	**124 Completed Tickets** mapped 1-to-1 with all 124 Git Commits
Future Work (To Do):	**14 Pre-Structured Tickets** for Big Data Lakehouse & Kafka Streaming
Total Story Points:	**579 Total Story Points** across 6 Sprints | 5 Team Members

1. 6-Sprint Agile Execution Roadmap & Velocity Distribution

Sprint	Theme / Scope	Tickets	Status	Story Points
Sprint 1	Architecture, Taxonomy & Ingest	30 Tickets	Done (100%)	**102 SP**
Sprint 2	Cost-Sensitive Triage & Platt Calibration	25 Tickets	Done (100%)	**95 SP**
Sprint 3	Causal Graph & Workshop Verifier	28 Tickets	Done (100%)	**106 SP**
Sprint 4	Dataset Scaling, Causal LM from Scratch & UI	41 Tickets	Done (100%)	**154 SP**
Sprint 5	[FUTURE] Kafka Ingestion & Parquet Lakehouse	6 Tickets	To Do (Reserved)	**34 SP**
Sprint 6	[FUTURE] PySpark Streaming, DuckDB & Defense	8 Tickets	To Do (Reserved)	**44 SP**
Total	**Full Platform Delivery**	**138 Tickets**	**124 Done / 14 To Do**	**579 SP**

2. Complete 138-Ticket Jira Catalog & Git Commit Traceability

Key	Sprint	Summary & Commit Tag	SP	Assignee	Status
AUT-095	Sprint	**Initial commit: AutoCare AI prototype with Stories 1-4, 12, RAG core, and React UI** [76b7256] Initial commit: AutoCare AI prototype with Stories 1-4,	5	Muhammad	Done
AUT-096	Sprint	**docs: add FYP proposal and user stories specifications** [3b3e2c1] docs: add FYP proposal and user stories specifications	3	Muhammad	Done
AUT-097	Sprint	**feat: complete FYP-1 implementation for 3 cars (2015-2025), intelligent modules M3-M9, and UI screen** [859693e] feat: complete FYP-1 implementation for 3 cars (2015-20	5	Muhammad	Done
AUT-098	Sprint	**fix: make user_vehicle_id optional on verification/case bank APIs and add graceful ml fallback** [b19f061] fix: make user_vehicle_id optional on verification/case	3	Muhammad	Done
AUT-099	Sprint	**feat: add 45 comprehensive vehicle manuals and downloader script across Corolla, Civic, and Swift (2** [bc39793] feat: add 45 comprehensive vehicle manuals and download	3	Muhammad	Done
AUT-117	Sprint	**ci: add GitHub Actions CI workflow, PR template, Docker compose, and Jira 6-month sprint backlog** [37ede42] ci: [AUT-117] add GitHub Actions CI workflow, PR templa	3	Muhammad	Done
AUT-101	Sprint	**test(taxonomy): add vehicle configuration schema unit tests** [565e202] test(taxonomy): [AUT-101] add vehicle configuration sch	3	Talha	Done
AUT-101	Sprint	**feat(taxonomy): seed 96 vehicle configurations for Corolla, Civic, Swift (2015-2025)** [e602e60] feat(taxonomy): [AUT-101] seed 96 vehicle configuration	3	Talha	Done
AUT-102	Sprint	**feat(dtc): ingest SAE J2012 diagnostic trouble codes and symptoms** [7dde6bb] feat(dtc): [AUT-102] ingest SAE J2012 diagnostic troubl	3	Talha	Done
AUT-102	Sprint	**test(dtc): add assertions for DTC severity and possible cause mapping** [b225afc] test(dtc): [AUT-102] add assertions for DTC severity an	3	Talha	Done
AUT-77	Sprint	**Merge pull request #1 from marketbul/feature/us1-vehicle-taxonomy** [0c83883] Merge pull request #1 from marketbul/feature/us1-vehicl	3	Muhammad	Done
AUT-103	Sprint	**feat(ingest): structure-aware PDF chunker isolating warnings and tables** [91d5193] feat(ingest): [AUT-103] structure-aware PDF chunker iso	3	Danial	Done
AUT-104	Sprint	**feat(retrieval): implement BM25 lexical retriever with tokenization** [977b502] feat(retrieval): [AUT-104] implement BM25 lexical retri	3	Danial	Done
AUT-105	Sprint	**feat(nlp): add Urdu and Roman Urdu automotive glossary translator** [cf7724f] feat(nlp): [AUT-105] add Urdu and Roman Urdu automotive	3	Danial	Done
AUT-104	Sprint	**test(retrieval): test hybrid BM25 + dense semantic retrieval on manual passages** [4d4eb42] test(retrieval): [AUT-104] test hybrid BM25 + dense sem	3	Danial	Done
AUT-82	Sprint	**Merge pull request #2 from marketbul/feature/us2-multilingual-rag** [61b91b9] Merge pull request #2 from marketbul/feature/us2-multil	5	Muhammad	Done
AUT-106	Sprint	**feat(ocr): camera fault code reader with regex verification** [d7243f0] feat(ocr): [AUT-106] camera fault code reader with rege	3	Danial	Done
AUT-106B	Sprint	**fix(ocr): enhance contrast and binarization for low-light scanner screens** [de41956] fix(ocr): [AUT-106B] enhance contrast and binarization	3	Danial	Done
AUT-85	Sprint	**Merge pull request #3 from marketbul/feature/us3-photo-code-scanner** [efd5dbf] Merge pull request #3 from marketbul/feature/us3-photo-	3	Muhammad	Done
AUT-107	Sprint	**test(triage): add baseline evaluation test on symptom corpora** [dc826eb] test(triage): [AUT-107] add baseline evaluation test on	5	Ahmad	Done
AUT-107	Sprint	**feat(ml): train baseline gradient boosting classifier (high Critical error 68.2%)** [4a2eb79] feat(ml): [AUT-107] train baseline gradient boosting cl	3	Ahmad	Done
AUT-107B	Sprint	**refactor(ml): implement cost-sensitive loss weighting dropping Critical FNR to 11.4%** [758b744] refactor(ml): [AUT-107B] implement cost-sensitive loss	3	Ahmad	Done
AUT-89	Sprint	**Merge pull request #4 from marketbul/feature/us4-cost-sensitive-triage** [20afd2b] Merge pull request #4 from marketbul/feature/us4-cost-s	5	Muhammad	Done
AUT-108	Sprint	**feat(calibration): implement Platt scaling and temperature calibration (M3)** [e410822] feat(calibration): [AUT-108] implement Platt scaling an	5	Ahmad	Done
AUT-108	Sprint	**test(calibration): evaluate Expected Calibration Error (ECE) and Brier reliability score** [ec2ff87] test(calibration): [AUT-108] evaluate Expected Calibrat	5	Ahmad	Done
AUT-109	Sprint	**feat(gate): add selective abstention gate for high-uncertainty vehicle queries** [bc33b8b] feat(gate): [AUT-109] add selective abstention gate for	3	Ahmad	Done
AUT-93	Sprint	**Merge pull request #5 from marketbul/feature/us5-calibrated-confidence** [0ced1cc] Merge pull request #5 from marketbul/feature/us5-calibr	3	Muhammad	Done
AUT-111	Sprint	**feat(trace): assemble transparent multi-hop explanation traces (M4)** [178ab5b] feat(trace): [AUT-111] assemble transparent multi-hop e	5	Talha	Done
AUT-111B	Sprint	**test(trace): verify explanation trace completeness across all 3 vehicles** [fb812a3] test(trace): [AUT-111B] verify explanation trace comple	3	Talha	Done

AUT-96	Sprint	**Merge pull request #6 from marketbul/feature/us6-explanation-trace** [d655210] Merge pull request #6 from marketbul/feature/us6-explan	**5**	Muhammad	**Done**
AUT-112	Sprint	**test(verifier): add failing unit test for workshop repair claim verification** [55d6f79] test(verifier): [AUT-112] add failing unit test for wor	**5**	Ahmad	**Done**
AUT-112	Sprint	**feat(verifier): implement repair verifier using premise entailment logic (M5)** [1bc023c] feat(verifier): [AUT-112] implement repair verifier usi	**5**	Ahmad	**Done**
AUT-112B	Sprint	**docs(verifier): enforce Proposal Section 17 non-accusatory ethical guidelines** [6b4261d] docs(verifier): [AUT-112B] enforce Proposal Section 17	**5**	Ahmad	**Done**
AUT-100	Sprint	**Merge pull request #7 from marketbul/feature/us7-repair-verifier** [2c9883b] Merge pull request #7 from marketbul/feature/us7-repair	**5**	Muhammad	**Done**
AUT-110	Sprint	**feat(kg): build NetworkX directed causal graph with 49 nodes and 36 causal edges (M6)** [d3fe41e] feat(kg): [AUT-110] build NetworkX directed causal grap	**5**	Talha	**Done**
AUT-110	Sprint	**test(kg): test multi-hop causal inference from overheating symptom to head gasket** [8be24e1] test(kg): [AUT-110] test multi-hop causal inference fro	**5**	Talha	**Done**
AUT-110	Sprint	**refactor(kg): optimize cycle detection and confidence propagation in causal chains** [09fe894] refactor(kg): [AUT-110] optimize cycle detection and co	**5**	Talha	**Done**
AUT-104	Sprint	**Merge pull request #8 from marketbul/feature/us8-knowledge-graph** [0a50519] Merge pull request #8 from marketbul/feature/us8-knowle	**3**	Muhammad	**Done**
AUT-114	Sprint	**feat(casebank): implement similar-case memory bank with resolution tracking (M8)** [70a0333] feat(casebank): [AUT-114] implement similar-case memory	**3**	Danial	**Done**
AUT-114	Sprint	**feat(casebank): add provenance tracking distinguishing seeded vs real user cases** [9183870] feat(casebank): [AUT-114] add provenance tracking disti	**3**	Danial	**Done**
AUT-107	Sprint	**Merge pull request #9 from marketbul/feature/us9-case-bank** [1dbaa7b] Merge pull request #9 from marketbul/feature/us9-case-b	**3**	Muhammad	**Done**
AUT-113	Sprint	**feat(health): calculate 6-component subsystem health index and mileage degradation (M7)** [d81ae36] feat(health): [AUT-113] calculate 6-component subsystem	**3**	Muhammad	**Done**
AUT-113	Sprint	**feat(health): implement scheduled maintenance forecaster with Pakistan severe climate multiplier** [27d4c25] feat(health): [AUT-113] implement scheduled maintenance	**3**	Muhammad	**Done**
AUT-113	Sprint	**test(health): verify maintenance interval prediction under severe city driving profile** [2cd6169] test(health): [AUT-113] verify maintenance interval pre	**3**	Muhammad	**Done**
AUT-111	Sprint	**Merge pull request #10 from marketbul/feature/us10-health-predictor** [b8909fb] Merge pull request #10 from marketbul/feature/us10-heal	**3**	Muhammad	**Done**
AUT-115	Sprint	**feat(report): implement ReportLab PDF service audit report generator (M9)** [97aa1ae] feat(report): [AUT-115] implement ReportLab PDF service	**3**	Muhammad	**Done**
AUT-115	Sprint	**test(report): test PDF certificate generation with case metadata and checklist** [214fcef] test(report): [AUT-115] test PDF certificate generation	**3**	Muhammad	**Done**
AUT-114	Sprint	**Merge pull request #11 from marketbul/feature/us11-audit-report-pdf** [9a40892] Merge pull request #11 from marketbul/feature/us11-audi	**3**	Muhammad	**Done**
AUT-116	Sprint	**feat(ui): add responsive React screens for repair verification and health index** [1006455] feat(ui): [AUT-116] add responsive React screens for re	**3**	Muhammad	**Done**
AUT-116	Sprint	**feat(ui): integrate SimilarCasesScreen and PDF report download in frontend** [cea7a3b] feat(ui): [AUT-116] integrate SimilarCasesScreen and PD	**3**	Muhammad	**Done**
AUT-119	Sprint	**feat(manuals): download and index 45 comprehensive OEM manuals for 3 cars** [19437e2] feat(manuals): [AUT-119] download and index 45 comprehe	**3**	Muhammad	**Done**
AUT-120	Sprint	**test(suite): author comprehensive acceptance test suite with 115 passing tests** [a1275ae] test(suite): [AUT-120] author comprehensive acceptance	**5**	Muhammad	**Done**
AUT-119	Sprint	**Merge pull request #12 from marketbul/feature/us12-vision-damage-assessor** [b4dd9aa] Merge pull request #12 from marketbul/feature/us12-visi	**3**	Muhammad	**Done**
AUT-120	Sprint	**chore(release): v1.0.0-fyp1 final integrated decision support system** [428ac44] chore(release): v1.0.0-fyp1 final integrated decision s	**3**	Muhammad	**Done**
AUT-122	Sprint	**fix(ci): resolve GitHub Actions CI build issues, add missing dependencies, and fix JSX comment synta** [ad8333f] fix(ci): [AUT-122] resolve GitHub Actions CI build issu	**5**	Muhammad	**Done**
AUT-122	Sprint	**fix(sql): change comment prefix to standard SQL double-dash in schema.sqlite.sql** [f49e4a3] fix(sql): change comment prefix to standard SQL double-	**3**	Muhammad	**Done**
AUT-123	Sprint	**fix(rag): add graceful fallback to extractive generator when no cloud LLM key is configured** [17f3d08] fix(rag): add graceful fallback to extractive generator	**5**	Muhammad	**Done**
AUT-123	Sprint	**docs: update project team members and add collaborator attribution** [24b4e07] docs: [AUT-123] update project team members and add col	**3**	Muhammad	**Done**
AUT-117	Sprint	**feat(corpus): implement interactive 45 manuals corpus explorer and hybrid RAG details** [4bcfc7f] feat(corpus): [AUT-117] implement interactive 45 manual	**5**	Danial	**Done**

AUT-118	Sprint	**feat(graph): add automotive causal knowledge graph visual inspector** [1e7b895] feat(graph): [AUT-118] add automotive causal knowledge	**5**	Talha	**Done**
AUT-119	Sprint	**feat(analytics): add model reliability dashboard with Platt calibration and triage FNR matrix** [16d93f6] feat(analytics): [AUT-119] add model reliability dashbo	**5**	Ahmad	**Done**
AUT-120	Sprint	**feat(pricing): implement Pakistan spare parts & workshop labour cost estimator in PKR** [178762a] feat(pricing): [AUT-120] implement Pakistan spare parts	**3**	Muhammad	**Done**
AUT-121	Sprint	**feat(system): integrate all 5-person FYP-1 modules, navigation, and defense architecture specificati** [dce9a7a] feat(system): [AUT-121] integrate all 5-person FYP-1 mo	**3**	Muhammad	**Done**
AUT-130	Sprint	**docs: finalize FYP-1 defense specification with Friday-to-Monday sprint git audit** [8c740f7] docs: finalize FYP-1 defense specification with Friday-	**3**	Muhammad	**Done**
AUT-131	Sprint	**fix(api): clean up service attribute assignment syntax in graph_export** [af1d294] fix(api): clean up service attribute assignment syntax	**5**	Muhammad	**Done**
AUT-124	Sprint	**feat(auth): implement Google OAuth token verification and automatic account provisioning** [d3133d5] feat(auth): [AUT-124] implement Google OAuth token veri	**5**	Talha	**Done**
AUT-124	Sprint	**test(auth): add comprehensive test suite for Google account auto-registration** [3868db6] test(auth): [AUT-124] add comprehensive test suite for	**5**	Talha	**Done**
AUT-124	Sprint	**feat(ui): integrate Google Sign-In button and instant verification modal in AuthScreen** [35c1736] feat(ui): [AUT-124] integrate Google Sign-In button and	**5**	Talha	**Done**
AUT-125	Sprint	**feat(kg): enhance multi-hop causal inference path scoring and diagnostic ranking** [744f8f2] feat(kg): [AUT-125] enhance multi-hop causal inference	**5**	Talha	**Done**
AUT-126	Sprint	**feat(ui): redesign web dashboard into Linear/Raycast dark-zinc Bento Grid layout** [9629858] feat(ui): [AUT-126] redesign web dashboard into Linear/	**5**	Talha	**Done**
AUT-127	Sprint	**feat(mobile): build cross-platform mobile apps using Flutter and React Native Expo** [3d4cbc5] feat(mobile): [AUT-127] build cross-platform mobile app	**5**	Muhammad	**Done**
AUT-128	Sprint	**docs(design): add Figma token studio tokens and design system specifications** [6468a89] docs(design): [AUT-128] add Figma token studio tokens a	**5**	Talha	**Done**
AUT-129	Sprint	**fix(api): robust attribute access for knowledge graph in export endpoint** [18a51f0] fix(api): [AUT-129] robust attribute access for knowled	**3**	Talha	**Done**
AUT-130	Sprint	**style(ui): aesthetic Linear and Raycast dark-zinc redesign with electric violet and emerald accents** [0e89971] style(ui): [AUT-130] aesthetic Linear and Raycast dark-	**5**	Talha	**Done**
AUT-131	Sprint	**feat(ui): clean dual-theme light/dark mode and simple SaaS landing card with role switcher** [4ff82a8] feat(ui): [AUT-131] clean dual-theme light/dark mode an	**5**	Muhammad	**Done**
AUT-132	Sprint	**feat(ui): add settings screen, simplify executive dashboard, and declutter navigation** [862cee4] feat(ui): [AUT-132] add settings screen, simplify execu	**5**	Muhammad	**Done**
AUT-133	Sprint	**refactor(ui): remove redundant theme toggle from lower left corner account bar** [7b2d407] refactor(ui): [AUT-133] remove redundant theme toggle f	**5**	Muhammad	**Done**
AUT-134	Sprint	**docs(business): Barringer & Ireland business model template for AutoCare AI (Chapter 4)** [1c676eb] docs(business): [AUT-134] Barringer & Ireland business	**3**	Muhammad	**Done**
AUT-135	Sprint	**docs(assignment): generate printable PDF assignment with all group members and roll numbers** [fe9ef56] docs(assignment): [AUT-135] generate printable PDF assi	**3**	Muhammad	**Done**
AUT-146	Sprint	**docs: update group member roll numbers and student names in business model assignment** [d26d061] docs: update group member roll numbers and student name	**3**	Abdullah	**Done**
AUT-147	Sprint	**docs: add AGENTS.md with permanent team roster and attribution guidelines** [ac69679] docs: add AGENTS.md with permanent team roster and attr	**5**	Abdullah	**Done**
AUT-148	Sprint	**docs: streamline business model pdf to clean 3-page template matching Chapter 4 rubric** [49f89d9] docs: streamline business model pdf to clean 3-page tem	**5**	Abdullah	**Done**
AUT-136	Sprint	**feat(ui): upgrade to enterprise SaaS UI/UX with sleek topbar, authentic Google OAuth, and Linear des** [f736ee5] feat(ui): [AUT-136] upgrade to enterprise SaaS UI/UX wi	**5**	Abdullah	**Done**
AUT-137	Sprint	**feat(ui): consolidate car owner diagnostic journey into unified 4-stage dashboard with PKR cost tria** [3a15e7c] feat(ui): [AUT-137] consolidate car owner diagnostic jo	**5**	Abdullah	**Done**
AUT-151	Sprint	**docs(pid): add 4-week professional term project initiation document in PDF and DOCX formats** [7d6535a] docs(pid): add 4-week professional term project initiat	**3**	Abdullah	**Done**
AUT-140	Sprint	**feat(arch): initialize AI Auto Pakistan data engine architecture, source registry, and audit logger** [981835b] feat(arch): [AUT-140] initialize AI Auto Pakistan data	**3**	Muhammad	**Done**

AUT-141	Sprint	**feat(taxonomy): implement 18-category Pakistani automotive fault taxonomy and fleet specs (Phase 2)** [09d09af] feat(taxonomy): [AUT-141] implement 18-category Pakista	**3**	Talha	**Done**
AUT-142	Sprint	**feat(schema): build normalized Pydantic automotive records and unit tests (Phase 2)** [6adc549] feat(schema): [AUT-142] build normalized Pydantic autom	**5**	Danial	**Done**
AUT-143	Sprint	**feat(sources): add YouTube collector, Facebook import, PakWheels importer, and consent-gated intervi** [d546df0] feat(sources): [AUT-143] add YouTube collector, Faceboo	**5**	Danial	**Done**
AUT-144	Sprint	**feat(templates): add CSV templates for Facebook, PakWheels, mechanic interviews, and owner reports (** [7e77834] feat(templates): [AUT-144] add CSV templates for Facebo	**3**	Muhammad	**Done**
AUT-145	Sprint	**feat(cleaning): implement canonical deduplication, multilingual normalizer, and quality validator (P** [0a2616b] feat(cleaning): [AUT-145] implement canonical deduplica	**5**	Ahmad	**Done**
AUT-146	Sprint	**feat(media): implement media asset vault and transcript claim extraction pipeline (Phase 8)** [65cf36b] feat(media): [AUT-146] implement media asset vault and	**5**	Danial	**Done**
AUT-147	Sprint	**feat(storage): add unified relational DDL schema and multi-source database manager (Phase 9)** [6066265] feat(storage): [AUT-147] add unified relational DDL sch	**5**	Muhammad	**Done**
AUT-148	Sprint	**feat(ml): build leak-free grouped ML dataset splitter with dataset card generation (Phase 11)** [8c11d5f] feat(ml): [AUT-148] build leak-free grouped ML dataset	**5**	Ahmad	**Done**
AUT-149	Sprint	**feat(reports): add dataset quality report generator, CLI import commands, and Phase 8-11 integration** [f239a27] feat(reports): [AUT-149] add dataset quality report gen	**3**	Muhammad	**Done**
AUT-150	Sprint	**chore(git): ignore dynamic runtime data engine outputs and log files** [46f02a6] chore(git): [AUT-150] ignore dynamic runtime data engin	**3**	Muhammad	**Done**
AUT-151	Sprint	**feat(storage): extend database manager with vehicle fleet seeding, review workflow, and router mount** [dcc632e] feat(storage): [AUT-151] extend database manager with v	**3**	Muhammad	**Done**
AUT-152	Sprint	**feat(ingest): add LocalDatasetConsolidator unifying OBD-II codes, symptom corpus, manuals, and PKR p** [cc111a7] feat(ingest): [AUT-152] add LocalDatasetConsolidator un	**3**	Danial	**Done**
AUT-153	Sprint	**test(server): isolate semantic symptom groups in ML splitter and add dashboard API integration tests** [9196192] test(server): [AUT-153] isolate semantic symptom groups	**5**	Ahmad	**Done**
AUT-154	Sprint	**feat(dashboard): build FastAPI dataset engineering API and interactive localhost dashboard** [0090282] feat(dashboard): [AUT-154] build FastAPI dataset engine	**5**	Muhammad	**Done**
AUT-155	Sprint	**feat(sources): add AutomotiveBlogImporter and CSV research template for authorized blogs** [510f5ca] feat(sources): [AUT-155] add AutomotiveBlogImporter and	**3**	Danial	**Done**
AUT-156	Sprint	**feat(dashboard): add blog CSV import, paste-CSV input, and in-app collection guide to localhost UI** [175f55c] feat(dashboard): [AUT-156] add blog CSV import, paste-C	**5**	Muhammad	**Done**
AUT-157	Sprint	**feat(youtube-media): add YouTube comment threads/solution extractor and ffmpeg authorized video/scre** [54513b4] feat(youtube-media): [AUT-157] add YouTube comment thre	**5**	Danial	**Done**
AUT-158	Sprint	**feat(ml-export): add AI-Agent JSONL exporter with verified-fact boundary and integration tests** [0f97bf8] feat(ml-export): [AUT-158] add AI-Agent JSONL exporter	**3**	Talha	**Done**
AUT-159	Sprint	**feat(pipeline): wire autonomous YouTube comments, authorized video processing, and agent-jsonl expor** [8ed39f4] feat(pipeline): [AUT-159] wire autonomous YouTube comme	**5**	Muhammad	**Done**
AUT-160	Sprint	**feat(media-damage): detect spoken Urdu/English damage cues and capture exact timestamp screenshots** [3ef1e02] feat(media-damage): [AUT-160] detect spoken Urdu/Englis	**5**	Talha	**Done**
AUT-161	Sprint	**feat(bundle): build zero-port standalone dataset folder exporter and offline file:// HTML viewer** [201811b] feat(bundle): [AUT-161] build zero-port standalone data	**5**	Ahmad	**Done**
AUT-162	Sprint	**feat(launcher): add root-level one-click run_auto_collector.bat and Consented_Mechanic_Videos drop f** [b5fa48e] feat(launcher): [AUT-162] add root-level one-click run_	**3**	Muhammad	**Done**
AUT-163	Sprint	**feat(cc-youtube): support Creative Commons videoLicense search and safe yt_dlp audio+video extractio** [23a19bc] feat(cc-youtube): [AUT-163] support Creative Commons vi	**5**	Danial	**Done**

AUT-164	Sprint	**fix(whisper): decode 16kHz WAV via stdlib wave+numpy for faster-whisper and link media_assets FKs** [f8f8f19] fix(whisper): [AUT-164] decode 16kHz WAV via stdlib wav	**5**	Talha	**Done**
AUT-165	Sprint	**feat(viewer): render screenshot/audio links in offline dataset_viewer.html and update stats test** [9bd5288] feat(viewer): [AUT-165] render screenshot/audio links i	**3**	Muhammad	**Done**
AUT-166	Sprint	**fix(youtube-audit): deduplicate CC-BY YouTube records, filter STT noise, and add fleet word-boundary** [8af0f39] fix(youtube-audit): [AUT-166] deduplicate CC-BY YouTube	**5**	Abdullah	**Done**
AUT-167	Sprint	**feat(youtube-qa): add Toyota/Suzuki/Honda queries, YouTube pagination, and consolidated Q&A; JSON exp** [6d7ffd3] feat(youtube-qa): [AUT-167] add Toyota/Suzuki/Honda que	**5**	Abdullah	**Done**
AUT-168	Sprint	**feat(facebook-local): add local Facebook_Community_QA drop folder, flexible Q&A; importer, and facebo** [0d50c32] feat(facebook-local): [AUT-168] add local Facebook_Comm	**5**	Abdullah	**Done**
AUT-169	Sprint	**feat(open-qa): add Stack Exchange Mechanics (CC-BY-SA) and NHTSA Safety REST collectors for RAG & ML** [7f630bf] feat(open-qa): [AUT-169] add Stack Exchange Mechanics (	**5**	Abdullah	**Done**
AUT-170	Sprint	**fix(open-qa): expand Stack Exchange/NHTSA queries, unify KIA brand casing, and disambiguate NHTSA co** [a0cb72f] fix(open-qa): [AUT-170] expand Stack Exchange/NHTSA que	**3**	Abdullah	**Done**
AUT-171	Sprint	**feat(scale-10k): scale open APIs, YouTube/PakWheels channel queries, and export caps to 10,760 recor** [32e2460] feat(scale-10k): [AUT-171] scale open APIs, YouTube/Pak	**5**	Abdullah	**Done**
AUT-184	Sprint	**feat(dataset): consolidate multi-source automotive dataset and automotive dataset engineer chat hist** [e7ce84a] feat(dataset): consolidate multi-source automotive data	**3**	Abdullah	**Done**
AUT-185	Sprint	**feat(rag): implement from-scratch RAG pipeline with custom tokenizer, dense vector embedder, hybrid** [aba6dff] feat(rag): implement from-scratch RAG pipeline with cus	**5**	Abdullah	**Done**
AUT-186	Sprint	**feat(rag): train causal transformer LM on automotive corpus and wire trained weights into RAG pipeli** [58b39bb] feat(rag): train causal transformer LM on automotive co	**5**	Abdullah	**Done**
AUT-187	Sprint	**feat(ui): elevate car owner experience with automotive diagnostic intelligence, Pakistani repair mar** [8613cf1] feat(ui): elevate car owner experience with automotive	**5**	Abdullah	**Done**
AUT-188	Sprint	**feat(ui): redesign into modern narrative-driven automotive engineering web portal with spacious hero** [61bf981] feat(ui): redesign into modern narrative-driven automot	**5**	Abdullah	**Done**
AUT-101-132	Sprint	**feat(jira): complete Agile Scrum workflow, team WBS, and Jira backlog export** [d3e47ed] feat(jira): [AUT-101-132] complete Agile Scrum workflow	**3**	Abdullah	**Done**
AUT-101-143	Sprint	**feat(jira): update Agile Scrum workflow for 5 team members (32 SP each, 160 SP total)** [8b3275c] feat(jira): [AUT-101-143] update Agile Scrum workflow f	**3**	Abdullah	**Done**
AUT-201	Sprint	**Docker Compose Stack: Apache Kafka (KRaft), MinIO S3 & Distributed Bus** [RESERVED FUTURE BIG DATA WORK]	**8**	Talha	**To Do**
AUT-202	Sprint	**High-Velocity OBD-II CAN-Bus Live Telemetry Generator (10,000 msg/s)** [RESERVED FUTURE BIG DATA WORK]	**5**	Talha	**To Do**
AUT-203	Sprint	**PyArrow Snappy Parquet Lakehouse Partitioning (make/model/year/region)** [RESERVED FUTURE BIG DATA WORK]	**5**	Talha	**To Do**
AUT-204	Sprint	**Consistent Hash Ring with 160 Virtual Nodes per Physical Node** [RESERVED FUTURE BIG DATA WORK]	**8**	Abdullah	**To Do**
AUT-205	Sprint	**SHA-256 CAS Deduplication Engine & Merkle Audit Integrity Log** [RESERVED FUTURE BIG DATA WORK]	**5**	Abdullah	**To Do**
AUT-206	Sprint	**Argon2id Cryptographic Password & Authentication Security Upgrade** [RESERVED FUTURE BIG DATA WORK]	**3**	Abdullah	**To Do**
AUT-207	Sprint	**PySpark Distributed DataFrame Moving Averages & Outlier Pipelines** [RESERVED FUTURE BIG DATA WORK]	**5**	Muhammad	**To Do**
AUT-208	Sprint	**Spark Structured Streaming 1-Second Sliding Window Outlier Sink** [RESERVED FUTURE BIG DATA WORK]	**8**	Muhammad	**To Do**
AUT-209	Sprint	**Embedded DuckDB In-Memory OLAP Direct-Query Engine for Parquet** [RESERVED FUTURE BIG DATA WORK]	**5**	Danial	**To Do**
AUT-210	Sprint	**Isolation Forest & Rolling Z-Score Fleet Anomaly Detector** [RESERVED FUTURE BIG DATA WORK]	**5**	Ahmad	**To Do**
AUT-211	Sprint	**Big Data Operational Telemetry REST APIs (/api/bigdata/metrics)** [RESERVED FUTURE BIG DATA WORK]	**5**	Muhammad	**To Do**
AUT-212	Sprint	**React Big Data Operations Dashboard Component with Live Ticker** [RESERVED FUTURE BIG DATA WORK]	**8**	Muhammad	**To Do**

AUT-213	Sprint	**10 Empirical Big Data Benchmark Experiments & Latency Profiling** [RESERVED FUTURE BIG DATA WORK]	**8**	Danial	**To Do**
AUT-214	Sprint	**Final Big Data System Architecture, DFD Level 0-2 & Defense Slides** [RESERVED FUTURE BIG DATA WORK]	**5**	Ahmad	**To Do**